



CSC4160:

SQL Databases with MySQL

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Agenda

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Background/History

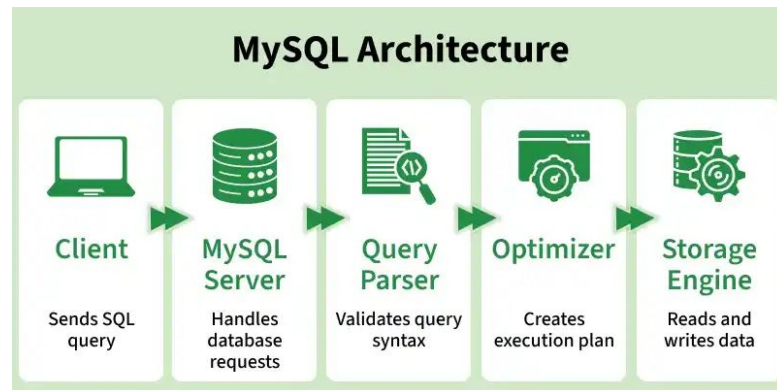
- SQL = Structured Query Language
 - Developed at IBM in the 1970s
- MySQL came later, developed and currently being maintained by Oracle
- Used to store and manage databases, using SQL to manage data





How MySQL Works

- MySQL Server: stores + manages databases
- MySQL Client: where you type in your commands
- Client sends SQL command
- Server checks and processes query
- Result returned to user





Benefits

- Strong security
 - User auth, SSL encryption, secure data storage options
- Scalability
- Free + open-source
- High performance
-



Demo

```
3      /* initialize database */
4      • CREATE DATABASE bakery;
5      • USE bakery;
6
7      /* initialize table */
8  ✓ • CREATE TABLE goods (
9          id INT PRIMARY KEY AUTO_INCREMENT,
10         name VARCHAR(100),
11         price INT
12     );
13
14     • SHOW TABLES;
```

Output

Result #5

Tables_in_bakery

goods



Demo

```
14      /* add values to table */
15      • INSERT INTO goods (name, price)
16        VALUES
17          ("croissant", 5),
18          ("chocolate chip cookie", 2),
19          ("bread loaf", 6);
20
21      /* display all info from table */
22      • SELECT * FROM goods;
```

Output

Result #6

id	name	price
1	croissant	5
2	chocolate chip cookie	2
3	bread loaf	6



Demo

```
21      /* filter thru table */
22      • SELECT id, name FROM goods
23      WHERE (price > 3);
```

Output

Result #6

Result #8

id

name

1

croissant

3

bread loaf



Demo

```
25      /* update value(s) */
26      • UPDATE goods
27        SET price = 2
28        WHERE name = "croissant";
29
30      • SELECT * FROM goods;
```

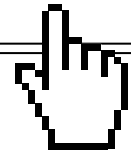
Output	Result #6	Result #8
id	name	price
1	croissant	2
2	chocolate chip cookie	2
3	bread loaf	6



Demo

```
32      /* delete value(s) */
33      • DELETE FROM goods
34        WHERE id = 1;
35
36      • SELECT * FROM goods;
```

Output	Result #6	Result #8	Result #10
id	name	price	
2	chocolate chip cookie	2	
3	bread loaf	6	



Summary

- Database management system maintained by Oracle
- Benefits including scalability and strong security
- Common CRUD operations include INSERT, SELECT, UPDATE, and DELETE



References

- <https://www.geeksforgeeks.org/mysql/what-is-mysql/>
- <https://www.geeksforgeeks.org/mysql/mysql-tutorial/>